VICTORY 2027

FORCE WORK POWER & ENERGY

LECTURE-06 PHYSICS

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Topics To Be Covered

1. REVISION
2. MECHANICAL ENERGY
3. WORK ENERGY THEOREM
4. QUESTIONS


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PE
- Position
- Configuration (Arrangement of molecules)

↓

+, 0, -

$ME = KE + PE$

Scalar Quantity



KE
- Linear Motion → Translational KE
- Rotational KE
- Vibrational KE

↓

always +ve

WORK ENERGY THEOREM



imp

When a force is applied in the direction of motion of a body, it accelerates the motion and thus increases the kinetic energy of the body. This increase in kinetic energy is equal to the work done by the force on the body. This is called the work-energy theorem.

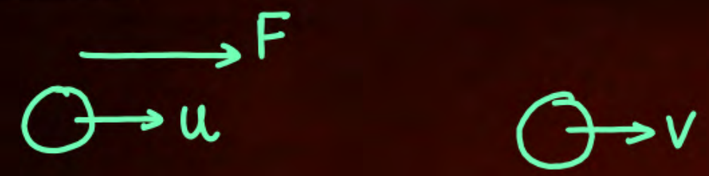
According to the work-energy theorem, the work done by a force on a moving body is equal to the increase in its kinetic energy.

$$W_{net} = \Delta KE = K_f - K_i$$

0 J

20 J

Derivation



$$v^2 = u^2 + 2as$$

$$v^2 - u^2 = 2as$$

$$\boxed{s = \frac{v^2 - u^2}{2a}}$$

Work Done

$$W = Fs$$

$$= (ma)\left(\frac{v^2 - u^2}{2a}\right)$$

$$\boxed{W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2}$$

$$\boxed{W = K_f - K_i}$$

WORK ENERGY THEOREM

Let a body of mass m be moving with an initial velocity u. When a constant force F is applied on the body along its direction of motion, it produces an acceleration a and the velocity of the body changes from u to v in moving a distance S. Then

From relation

$$v^2 = u^2 + 2aS$$

Displacement

$$S = \frac{v^2 - u^2}{2a}$$

$$W = F \times S = ma \times \left(\frac{v^2 - u^2}{2a}\right)$$

$$W = \frac{1}{2}m(v^2 - u^2)$$

$$W = \frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{mv^2} - \frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{mu^2}$$

$W = K_f - K_i =$ Increase in kinetic energy

QUESTIONS

Calculate the kinetic energy of a body of mass 2kg moving with a speed of 10ms^{-1}.

$$KE = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 2 \times 10^2$$

$$= 100\ \text{J}$$

QUESTIONS

How fast should a man weighing $600N$ run so that his kinetic energy is $750J$?
$(g = 10ms^{-2})$

$mg = 600$

$m \times 10 = 60\phi$

$\boxed{m = 60\ kg}$

$KE = \frac{1}{2}mv^2 = 750$

$\frac{1}{2} \times 60 \, v^2 = 75\phi$

$v^2 = 25$

$\boxed{v = 5\ m/s}$

QUESTIONS

How is the kinetic energy of a moving cart affected if (a) its mass is doubled, (b) its velocity is reduced to $\frac{1}{3}$rd of the initial velocity?

$K = \frac{1}{2}mv^2$

$K' = \frac{1}{2}m\left(\frac{v}{3}\right)^2$

$\boxed{K' = \frac{K}{9}}$

$K = \frac{1}{2}mv^2$

$K' = \frac{1}{2}(2m)v^2$

$\boxed{K' = 2K}$

$m' = 2m$

$V' = \frac{v}{3}$

$K = \frac{1}{2}mv^2$

$K' = \frac{1}{2}(2m)\left(\frac{v}{3}\right)^2$

$\boxed{K' = \frac{2}{9}K}$

FORMS OF KINETIC ENERGY

A body can have three types of motion namely the translational, rotational and vibrational, so the kinetic energy is also of three forms:

(1) translational kinetic energy,

(2) rotational kinetic energy, and

(3) vibrational kinetic energy.

TRANSLATIONAL KINETIC ENERGY

The motion of a body in a straight line path is called the translational motion and the kinetic energy of the body due to motion in a straight line is called the translational kinetic energy.

Examples: A car moving in a straight path, a freely falling body, a molecule of monoatomic gas have the translational kinetic energy.

He
Ne

ROTATIONAL KINETIC ENERGY

If a body rotates about an axis, the motion is called the rotational motion and the kinetic energy of the body due to rotational motion is called the rotational kinetic energy or simply the rotational energy.

Examples : A rotating wheel have the rotational kinetic energy. The earth rotating on its own axis has a huge amount of rotational kinetic energy. The atoms in a diatomic (or polyatomic") molecule rotate about a fixed axis in addition to their translational motion, so they possess both the rotational and translational kinetic energies.

N_2
O_2

NH_3
CO_2

VIBRATIONAL KINETIC ENERGY

If a body moves to and fro about its mean position, the motion is called the vibrational motion. The kinetic energy of the body due to its vibrational motion is called the vibrational kinetic energy or simply the vibrational energy.

Examples:

- A wire clamped at both the ends, when struck in the middle, vibrates.
- A steel strip, clamped at one end, vibrates when the other free end is displaced slightly and then released.
- In a solid, the atoms vibrate about their mean positions and so they possess the vibrational kinetic energy.
- In a polyatomic molecule the atoms can vibrate about their mean positions, so a polyatomic molecule has the vibrational energy in addition to the rotational and translational energies.

VIBRATIONAL KINETIC ENERGY

Note: Depending upon its state of motion, a moving body may possess either one or more than one form of kinetic energy simultaneously. For example, a rolling ball and the wheel of a running vehicle have both the translational and rotational kinetic energies simultaneously.

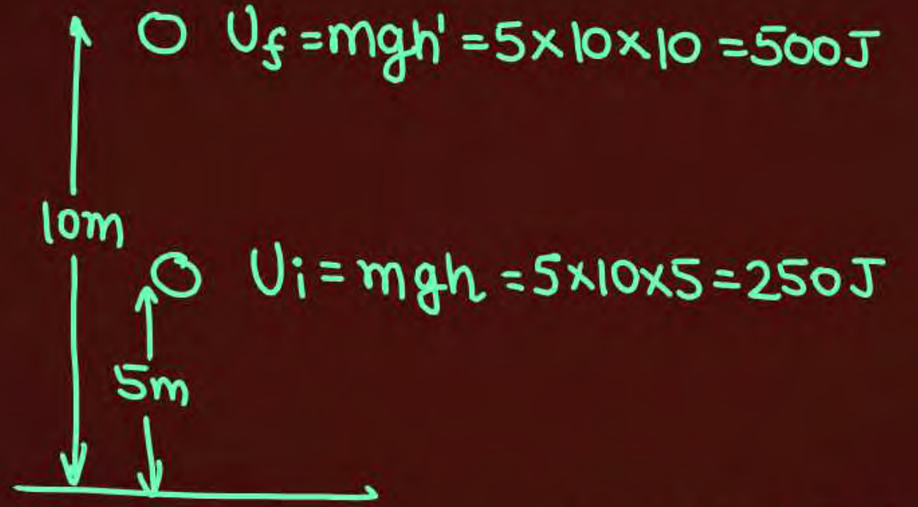
DISTINCTION BETWEEN THE POTENTIAL ENERGY AND KINETIC ENERGY

HW

	Potential Energy		Kinetic Energy
1.	It is the energy possessed by a body due to its changed position or configuration.	1.	It is the energy possessed by a body due to its state of motion.
2.	It is equal to the work done in bringing the body to its changed state.	2.	It is equal to the work that a moving body can do before coming to rest.
3.	It can change only in form of kinetic energy.	3.	It can change into any other form.
4.	it does not depend on the speed of the body.	4.	It depends on the speed of the body.

QUESTIONS

A body of mass 5kg is taken from a height 5m to 10m. Find the increase in its potential energy $(g = 10ms^{-2})$.



QUESTIONS

A body of mass 1kg falls from a height of 5m. How much energy does it possess at any instant? (Take $g = 9.8ms^{-2}$)

$U = mgh$

$= 1 \times 9.8 \times 5$

$U = 49\ J$

QUESTIONS

A block of mass 30kg is pulled up a slope as shown in Fig. with a constant speed by applying a force of 200N parallel to the slope from the initial position A to the final position B.

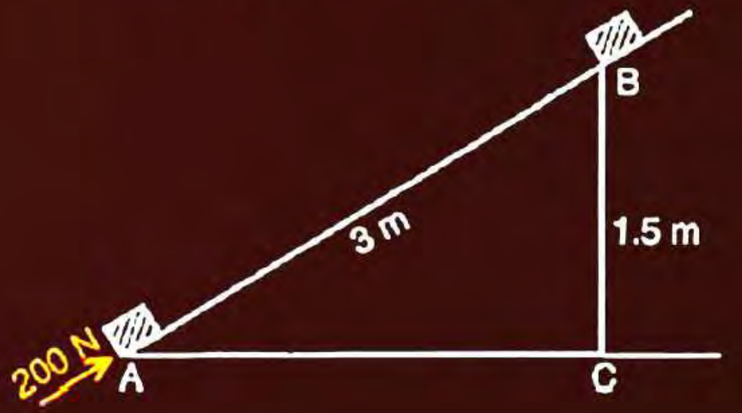
(a) Calculate:

(i) the work done by the force in moving the block from A to B, and $W = Fs = 200 \times 3 = 600\text{J}$

(ii) the potential energy gained by the block. Take : $g = 10\text{ms}^{-2}$ $U = mgh = 30 \times 10 \times 1.5 = 450\text{J}$

(b) Account for the difference in answers to parts (i) and (ii).

→ wasted against friction



QUESTIONS

In a dam, water falls at a rate of 1000kgs^{-1} from a height of 100m.

(a) Calculate the initial potential energy of the water.

(b) Assuming that 60% of the energy of the falling water is converted to electrical energy, calculate the power generated. (Take $g = 9.8ms^{-2}$).

(a) $U = mgh$

$= 1000 \times 9.8 \times 100$

$= 980 \times 1000$

$\boxed{U = 9.8 \times 10^5\ J}$

(b) $P = \frac{E}{t} = \frac{9.8 \times 10^5 \times \frac{60}{100}}{1} = 5.88 \times 10^5\ W$

QUESTIONS

What is the work to be done to increase the velocity of a car from 36kmh^{-1} to 54kmh^{-1}, if the mass of the car is 1500kg ?

$36 \times \frac{5}{18}$

$= 10\ \text{m/s}$

$54 \times \frac{5}{18}$

$= 15\ \text{m/s}.$

$1\ \text{Km/h} = \frac{5}{18}\ \text{m/s}.$

By Work-Energy Theorem

$W = K_f - K_i$

$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$= \frac{1}{2} \times 1500 (15)^2 - \frac{1}{2} \times 1500 \times (10)^2$

$= 750 \times 225 - 750 \times 100$

$= 750 (225 - 100)$

$W = 93750\ \text{J}$

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

A truck driver loads some oil drums into a truck by lifting them directly. Each drum has a mass of 80kg and the platform of the truck is at a height of 0.8m above the ground.

(a) What force is needed to lift a drum into the truck ?

(b) How much energy is used up in lifting a drum?

(c) After the truck is loaded, the driver drives off. List the major energy changes that take place in moving the truck.

(d) The driver stops the truck at the factory gate. What happens to the kinetic energy of the truck? Take $g = 10\text{ms}^{-2}$.

Solution: Given : $m = 80\text{kg}, h = 0.8\text{m}, g = 10\text{ms}^{-2}$

(a) Force needed to lift a drum

= Force of gravity on drum

$= mg = 80 \times 10 = 800\text{N}$

(b) Energy used up in lifting a drum

= gravitational potential energy mgh

$= 80 \times 10 \times 0.8 = 640\text{J}$

(c) In moving the truck, the chemical energy of the fuel (diesel) changes into the mechanical (kinetic) energy.

(d) On stopping the truck, the kinetic energy of the truck changes into the heat and sound energies.

QUESTIONS

2. A truck weighing $5 \times 10^3 \text{kgf}$ and a cart weighing 500kgf are moving with the same speed. Compare their kinetic energies.

3. For the same kinetic energy of a body, what should be the change in its velocity if its mass is increased four times?

Thank You